

Definitions:

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Let $a \in G$. Let $\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$.

- What is $\langle -1 \rangle$ in $\mathbb{Z}$?
- What is $\langle R_{240} \rangle$ in D_3 ? What is $\langle F_V \rangle$ in D_3 ?

Theorem (One-Step Subgroup Test). Let G be a group and H nonempty subset of G .

Steps:

- 1.

- 2.

Example 1. Let a be an element in a group G . Use the one-step subgroup test to prove that $\langle a \rangle$ is a subgroup of G .

Theorem (Two-Step Subgroup Test). Let G be a group and H nonempty subset of G .

Steps:

- 1.
- 2.
- 3.

Example 2. Let a be an element in a group G . Use the two-step subgroup test to prove that $\langle a \rangle$ is a subgroup of G .

Theorem (Finite Subgroup Test). Let G be a group and H nonempty subset of G .

Definition. The *center*, $Z(G)$, of a group G is the:

Question. What is the center of D_n ?

Definition. The *centralizer of a in G* , $C(a)$, is the:

Exercise. Prove that if G is a group and $a \in G$, then $C(a)$ is a subgroup of G .

Question. Is $C(a)$ Abelian? Is $Z(G)$ Abelian?

Exercise. Suppose a belongs to a group and $|a| = 5$. Prove that $C(a) = C(a^3)$.

The above statement is true only because $|a| = 5$. Find an element a from a group with such that $|a| = 6$ and $C(a) \neq C(a^3)$.